

Student Guide

Endpoint Behavioral Analysis Simulation — Process & Persistence

What This Simulation Is Training You To Do

This simulation trains you to analyze **endpoint behavior over time**, not individual alerts in isolation. Your goal is to determine whether observed activity represents:

- Normal user or system behavior
- Suspicious but inconclusive activity
- Confirmed malicious execution
- Persistence being established

The key skill is **behavioral correlation** — understanding how processes relate to each other and what they imply when viewed as a chain.

The Analyst Workflow (Endpoint-Focused)

Step 1 — Start With the Process Tree, Not the Alert Name

Ignore the alert label at first. Instead, examine:

- Parent process
- Child process
- Execution context (user vs system)
- Timing between processes

Ask: “Does this process relationship make sense for a normal endpoint?”

For example, a browser spawning a helper process is normal. A document reader spawning a scripting engine is not.

Step 2 — Identify Execution Red Flags

Certain behaviors strongly indicate malicious execution, especially when combined:

- Office applications spawning PowerShell or cmd
- Encoded or hidden command-line arguments
- Use of system binaries for unintended purposes (living-off-the-land)
- Execution from user-writable directories (Temp, AppData, Downloads)

None of these alone guarantee malware, but **multiple indicators together** raise confidence.

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Step 3 — Separate Execution From Persistence

This is a critical distinction for both real SOC work and exams.

- **Execution** answers: “Did malicious code run?”
- **Persistence** answers: “Will it run again after reboot?”

Persistence mechanisms often include:

- Scheduled tasks
- Services
- Registry run keys
- Startup folder entries

If you only see execution with no persistence artifacts, the correct conclusion is **execution without confirmed persistence**, not “fully installed malware.”

Step 4 — Evaluate Timing and Sequencing

Order matters. Typical malicious flow:

1. User action or exploit triggers execution
2. Script or binary downloads or stages payload
3. Persistence is created shortly after

If persistence appears **minutes after suspicious execution**, that correlation strengthens your conclusion.

Step 5 — Watch for Legitimate Look-Alikes

This simulation intentionally includes behaviors that can occur legitimately:

- PowerShell used by IT automation
- Scheduled tasks created by updates or management tools
- Services installed by enterprise software

To differentiate:

- Look at naming conventions
- File locations
- Whether behavior aligns with known tooling in the environment

Malware often hides by **mimicking normal names** but uses abnormal paths or timing.

Step 6 — Choose the Most Defensible Conclusion

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Your final choice should reflect **what the evidence proves**, not the worst-case scenario.

Examples:

- “Suspicious execution observed, no persistence confirmed”
- “Persistence established via scheduled task and service creation”

Avoid over-classification — that’s a common analyst and exam mistake.

What This Simulation Is *Not* Asking You To Do

- Reverse engineer malware
- Attribute the attack to a threat actor
- Prove full compromise beyond observed evidence

You are operating at **Tier 1–2 SOC depth**, not incident response or forensics.

Analyst Mindset Reminder

Strong endpoint analysis comes from asking: “What behavior is unusual *for this endpoint at this time?*”

Not: “Does this look scary?”

Precision and restraint are marks of a skilled analyst.